DGMS 2019 • SECTION E • QUESTION 28

Coastal Hydrodynamics: Wave-Dominated Coastline Zonation from Land to Sea

Directorate General of Mines Safety (DGMS) Junior Scientific Officer Examination 2019 | Sedimentology & Coastal Processes Guide

Question Statement

Q.28. In a wave-dominated coastline, the correct distribution of environment from land to sea is:

a (a) Shoaling zone – surf zone – breaker zone – swash zone

b (b) Swash zone – shoaling zone – surf zone – breaker zone

c (c) Swash zone – surf zone – breaker zone – shoaling zone

d (d) Surf zone – shoaling zone – swash zone – breaker zone

Correct Answer: (c) Swash zone – surf zone – breaker zone – shoaling zone

LANDWARD (Beach) → Swash Zone → Surf Zone → Breaker Zone → Shoaling Zone → SEAWARD (Offshore)

Official Textbook Figures (Sam Boggs Jr., Principles of Sedimentology & Stratigraphy)

diameter downward in the water column. As deep-water orbital waves ap-
ɔach shallow water where depth is about one-half the wave length, the orbital
ɔtion of the water is impeded by interaction with the bottom. Orbits become
ɔgressively more elliptical and, eventually, near the bottom, develop a nearly
rizontal to-and-fro motion that can move sediment back and forth. This to-and-
movement is important in generating ripple bedforms as well as in producing
ne net sediment transport. As waves progress farther shoreward into the shal-
v **shoaling zone** (Fig. 9.21), forward velocity of the wave slows, wave length de-
ases, and wave height increases. The waves eventually steepen to the point
ıere orbital velocity exceeds wave velocity and the wave breaks, creating the

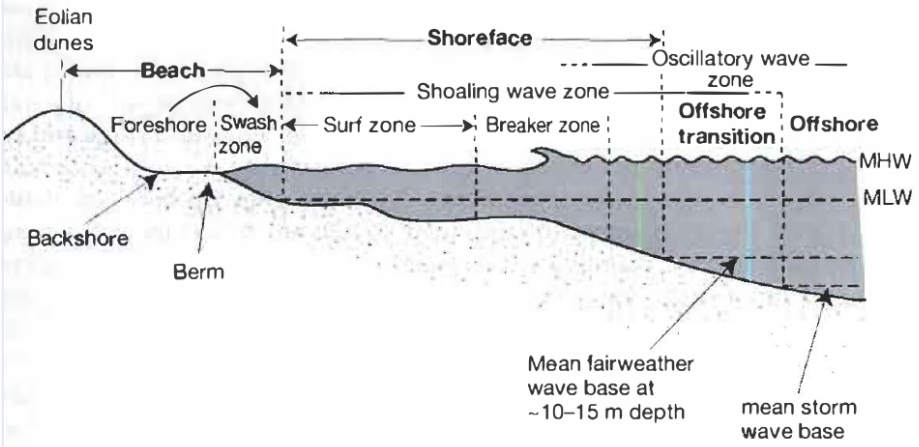
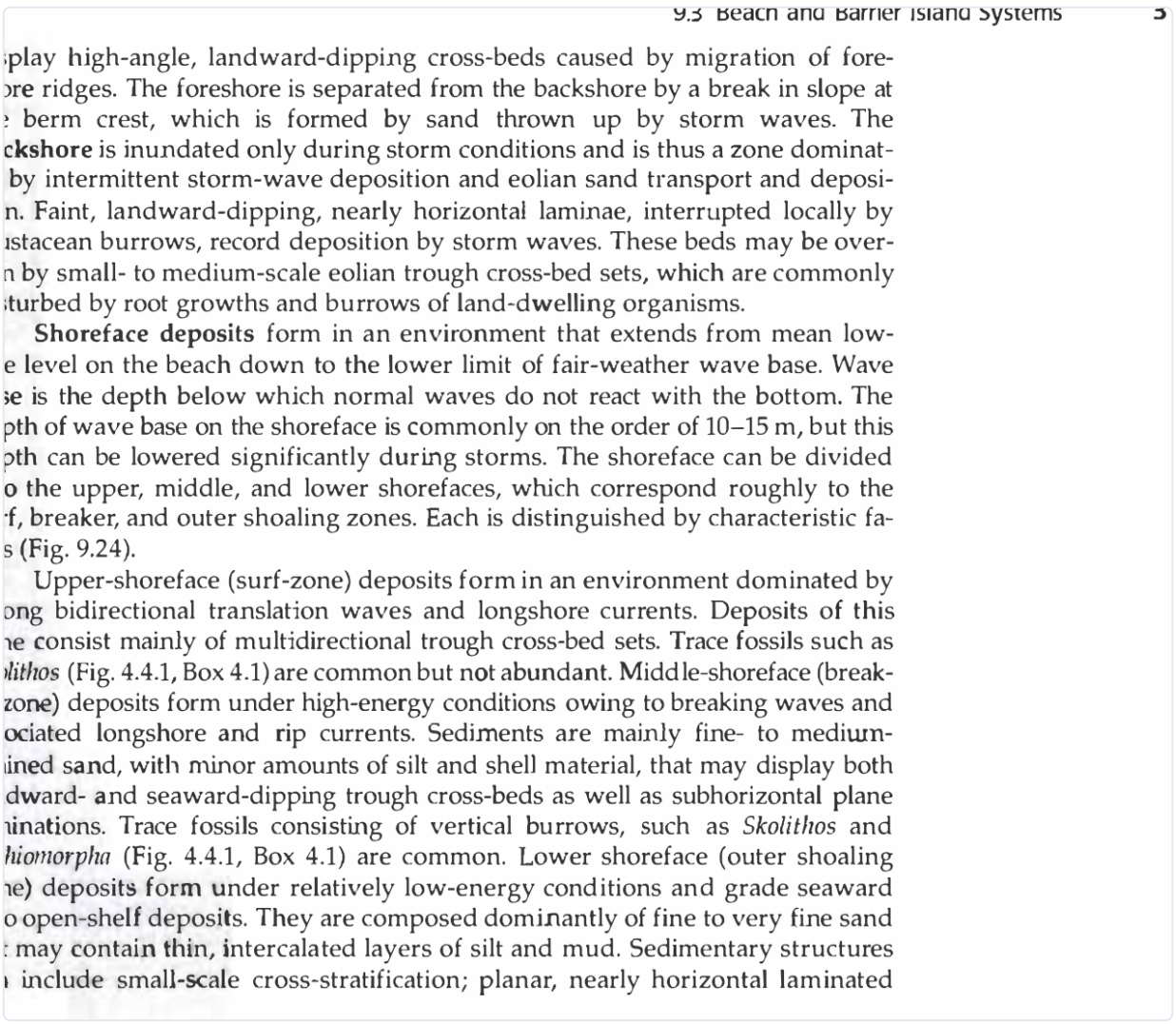


Figure 9.21
Generalized cross-sectional profile of the beach and nearshore zone, showing a the principal zones of wave activity.

Figure 1: Sam Boggs Jr. (4th/5th Ed.), Chapter 9, Figure 9.21. Generalized cross-sectional profile of the beach and nearshore zone, showing the principal zones of wave activity. Observe the seaward progression from the subaerial beach: **Foreshore** / **Swash Zone** (run-up on beach face) → **Surf Zone** (inner translation bores) → **Breaker Zone** (where waves break) → **Shoaling Zone** (where waves feel bottom in shoreface).



9.3 Beach and Barrier Island Systems 5

play high-angle, landward-dipping cross-beds caused by migration of fore-
ore ridges. The foreshore is separated from the backshore by a break in slope at
e berm crest, which is formed by sand thrown up by storm waves. The
ckshore is inundated only during storm conditions and is thus a zone dominat-
by intermittent storm-wave deposition and eolian sand transport and deposi-
n. Faint, landward-dipping, nearly horizontal laminae, interrupted locally by
ıstacean burrows, record deposition by storm waves. These beds may be over-
n by small- to medium-scale eolian trough cross-bed sets, which are commonly
turbed by root growths and burrows of land-dwelling organisms.

Shoreface deposits form in an environment that extends from mean low-
e level on the beach down to the lower limit of fair-weather wave base. Wave
se is the depth below which normal waves do not react with the bottom. The
pth of wave base on the shoreface is commonly on the order of 10–15 m, but this
pth can be lowered significantly during storms. The shoreface can be divided
o the upper, middle, and lower shorefaces, which correspond roughly to the
f, breaker, and outer shoaling zones. Each is distinguished by characteristic fa-
s (Fig. 9.24).

Upper-shoreface (surf-zone) deposits form in an environment dominated by
ong bidirectional translation waves and longshore currents. Deposits of this
ne consist mainly of multidirectional trough cross-bed sets. Trace fossils such as
olithos (Fig. 4.4.1, Box 4.1) are common but not abundant. Middle-shoreface (break-
zone) deposits form under high-energy conditions owing to breaking waves and
ociated longshore and rip currents. Sediments are mainly fine- to medium-
ined sand, with minor amounts of silt and shell material, that may display both
dward- and seaward-dipping trough cross-beds as well as subhorizontal plane
inations. Trace fossils consisting of vertical burrows, such as *Skolithos* and
hiomorpha (Fig. 4.4.1, Box 4.1) are common. Lower shoreface (outer shoaling
ne) deposits form under relatively low-energy conditions and grade seaward
o open-shelf deposits. They are composed dominantly of fine to very fine sand
may contain thin, intercalated layers of silt and mud. Sedimentary structures
include small-scale cross-stratification; planar, nearly horizontal laminated

Figure 2: Sam Boggs Jr., Chapter 9, Figure 9.24. Zonation of sedimentary structures and facies across the shoreface, nearshore, and foreshore.

In-Depth Option Evaluation & Spatial Logic

(c) Swash zone – surf zone – breaker zone – shoaling zone — [CORRECT]

Spatial Progression: This sequence accurately reflects the hydrodynamic and morphological zones encountered when traveling from the dry land (beach berm) outward into the open sea:

- **1. Swash Zone (Landward-most):** The intertidal beach face (foreshore) alternately covered by upswash (run-up) and backwash (gravity drainage).
- **2. Surf Zone:** The zone of high-velocity translatory bores moving landward between the breaker bar and the swash limit.

- **3. Breaker Zone:** The narrow band where steepening waves become dynamically unstable and break violently ($H/h \approx 0.78$).
- **4. Shoaling Zone (Seaward-most):** The shoreface region from fair-weather wave base ($10\text{–}15\text{ m}$) where oscillatory deep-water waves begin interacting with the seafloor.

(a) Shoaling zone – surf zone – breaker zone – swash zone — [INCORRECT DIRECTION / REVERSED TRAP]

This option represents the progression from **SEA TO LAND** (seaward to onshore), but it scrambles the position of the breaker and surf zones. Waves enter the shoaling zone first, break in the breaker zone, move as surf, and run up as swash. Crucially, the prompt requested **FROM LAND TO SEA**!

(b) Swash zone – shoaling zone – surf zone – breaker zone — [INCORRECT]

The shoaling zone is located out at sea (lower to middle shoreface). Placing it immediately adjacent to the swash zone completely violates physical coastal architecture.

(d) Surf zone – shoaling zone – swash zone – breaker zone — [INCORRECT]

Completely disorganized hydrodynamic order; swash zone is landward of surf zone, and shoaling zone is outermost.

Hydrodynamic & Sedimentological Characteristics of Coastal Zones

Zone (Land → Sea)	Morphological Sub-environment	Dominant Hydrodynamic Process	Diagnostic Sedimentary Structures	Grain Size & Sorting
1. Swash Zone (Landward)	Foreshore / Beach Face	Alternating upswash and backwash flows; thin sheet flows	Low-angle planar lamination (dipping 2°–8° seaward), heavy-mineral placer laminae	Medium to coarse sand, exceptionally well-sorted
2. Surf Zone	Upper Shoreface / Nearshore Trough	Translational bores, longshore currents, rip current generation	Trough & planar cross-bedding, multidirectional wave ripples, subaqueous bar bedforms	Fine to coarse sand, moderately to well-sorted
3. Breaker Zone	Middle Shoreface / Breaker Bar Crest	Violent wave collapse, maximum turbulent kinetic energy, high bed shear	Coarse lag deposits, steep rip-channel scours, high-angle onshore-dipping cross-beds	Coarse sand, granules, shells; poorly sorted lag
4. Shoaling Zone (Seaward)	Lower Shoreface (Fair-weather wave base)	Symmetrical & asymmetrical oscillatory bottom currents ("feeling the bottom")	Symmetrical wave ripples, hummocky cross-stratification (HCS), *Ophiomorpha* burrows	Very fine to fine sand grading seaward to silt/ mud

Direct Textbook Quotations & Citations

Sam Boggs Jr. (2006), *Principles of Sedimentology and Stratigraphy, 4th ed., Chapter 9, pp. 324–325:*

"As waves progress farther shoreward into the shallow **shoaling zone** *(Fig. 9.21), forward velocity of the wave slows, wave length decreases, and wave height increases. The waves eventually steepen and become unstable, collapsing in the* **breaker zone***... In the* **surf zone***, a high-velocity translation wave (bore) is projected up the upper shoreface... At the shoreline, the surf zone gives way to the* **swash zone***, in which a rapid, very shallow swash flow moves up the beach."*

— Sam Boggs Jr., Chapter 9, pp. 324–325

H. G. Reading (1996), *Sedimentary Environments: Processes, Facies and Stratigraphy, 3rd ed., Chapter 8:*

"The beach profile in cross-section from the subaerial barrier or mainland to offshore comprises the **swash zone** *on the foreshore, passing seawards into the* **surf zone***, bounded externally by the* **breaker zone***, which in turn grades into the offshore* **shoaling wave zone***."*

— H. G. Reading (1996), Chapter 8

Exam Traps & Rapid Elimination Strategy

Essential Directional Rule for UPSC & DGMS Geology:

1. **Read the Direction Carefully:** Examiners love flipping between *"from land to sea"* and *"from sea to land"*.
   - **Land to Sea:** Swash ➔ Surf ➔ Breaker ➔ Shoaling.
   - **Sea to Land:** Shoaling ➔ Breaker ➔ Surf ➔ Swash.
2. **Remember the Beach Experience:** When you stand with your toes on the dry sand, water splashes your feet in the **Swash** zone. When you step into waist-deep water where waves are tumbling, you are in the **Surf** zone. Further out where surfers wait for waves to pitch and curl, that is the **Breaker** zone. Out where boats bob on gentle rolling swells, that is the **Shoaling** zone.